1

1.1

1.1.1

- **1803 Dalton**
- **1897 Thomson** $\frac{e}{m_e} = 1.758 \times 10^{11}\,\mathrm{C\,kg^{-1}}$
- **1909 Millikan** $e = 1.592 \times 10^{-19}\,\mathrm{C}$
- **1911 Rutherford** α

1.1.2

- **Rydberg** $\tilde{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
- $R = 1.097 \times 10^{7}\,\mathrm{m^{-1}}$
-

1.1.3

- $E = h\nu$ $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$
-

$$r = 52.9\,\mathrm{pm} \times \frac{n^2}{Z}$$
$$E = -21.8 \times 10^{-19}\,\mathrm{J} \times \frac{Z^2}{n^2}$$

- $\Delta E = E_2 - E_1 = h\nu$

1.1.4

- $\lambda = \frac{h}{p} = \frac{h}{mv}$
- $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$
- -

1.2

1.2.1

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

1.2.2

- n
- l $|M| = \frac{h}{2\pi}\sqrt{l(l+1)}$
- m $m = 0, \pm 1, \ldots, \pm l$
- m_s $m_s = \pm\frac{1}{2}$

1.2.3

- $|\psi|^2$
- $D(r) = 4\pi r^2|\psi|^2$
- $a_0 = 52.9\,\mathrm{pm}$

1.2.4

n				
1	$1s$	1	n^2	$2n^2$
2	$2s, 2p$	4	4	8
3	$3s, 3p, 3d$	9	9	18
4	$4s, 4p, 4d, 4f$	16	16	32

1.3

1.3.1

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-
-

1.3.2

1	$1s$	2
2	$2s, 2p$	8
3	$3s, 3p$	8
4	$4s, 3d, 4p$	18

1.3.3

s	I A II A	ns^{1-2}
p	III A∼VII A 0	ns^2np^{1-6}
d	III B∼VII B VIII	$(n-1)d^{1-8}ns^{1-2}$
ds	I B II B	$(n-1)d^{10}ns^{1-2}$
f		$(n-2)f^{0-14}(n-1)d^{0-1}ns^2$

1.4

1.4.1

- $Z' = Z - \sum \sigma$
-
  - K $Z' = 19 - 16.8 = 2.2$
  - Ca $Z' = 20 - 17.15 = 2.85$
  - Sc $Z' = 21 - 17.9 = 3.00$

1.4.2

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-
-
- $5 \rightarrow 6$

1.4.3

- $A(g) \longrightarrow A^{+}(g) + e^{-}$
- Pauling F (3.98) Cs (0.79)
- 2.0